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The field of embodied artificial intelligence (AI) is undergoing a paradigm shift, moving from perception-centric models to action-oriented systems capable of interacting with and manipulating the physical world. Central to this transformation is the emergence of Vision-Language-Action (VLA) models, which seek to unify visual perception, linguistic understanding, and motor control within a single, end-to-end learnable framework [5, 8, 9]. By leveraging the representational power of large language models (LLMs) and vision-language models (VLMs), VLA models have demonstrated remarkable proficiency in grounding natural language instructions into executable robotic actions, enabling tasks ranging from complex manipulation to long-horizon navigation [2, 3]. This integration promises to bridge the gap between high-level reasoning and low-level physical execution, a long-standing challenge in robotics.

Despite these advances, current VLA models face a fundamental limitation: they are predominantly designed to map observations directly to actions, often without an explicit, internalized understanding of the world's dynamics. While effective in controlled settings, this reactive paradigm struggles with generalization, robustness, and long-term planning. For instance, a VLA trained to grasp a cup may fail catastrophically if the cup is moved, the lighting changes, or an unexpected obstacle appears, as it lacks a predictive model of how the world evolves in response to its actions \cite{2602.17659v1}. This deficiency is particularly acute in real-world scenarios characterized by partial observability, stochasticity, and the need for multi-step reasoning. The challenge is further compounded by the computational demands of real-time operation \cite{2510.26742v1}, the difficulty of handling degraded sensory inputs such as dark or blurred scenes \cite{2604.04834v1}, and the scarcity of high-quality, diverse training data for tiny-scale models \cite{2509.09372v2}.

The core research gap, therefore, lies in the absence of a unified framework that seamlessly integrates action execution with a generative world model—a model that can simulate, predict, and reason about the consequences of actions before they are taken. Current approaches often treat world modeling and action policy as separate, disjoint modules, leading to inefficiencies and a lack of synergy. While some works have explored joint learning with motion image diffusion to improve visual representations \cite{2512.18007v1}, or have employed RL-augmented teleoperation for dexterous manipulation \cite{2603.08122v1}, these efforts remain piecemeal. A truly comprehensive paradigm must reconcile the need for precise, reactive control with the capacity for anticipatory, model-based reasoning. The concept of a "World Action Model" (WAM) emerges precisely to address this gap: a holistic architecture where the VLA policy is not merely a reactive controller but is co-designed with a world model that provides a rich, internal simulation of the environment's state transitions.

This survey presents a comprehensive review of the emerging paradigm of World Action Models, which we define as a unified framework for embodied AI that synthesizes Vision-Language-Action models with explicit world modeling capabilities. Our primary contribution is threefold. First, we propose a novel taxonomy that systematically categorizes existing research along two key dimensions: the degree of world model integration (from implicit to explicit) and the nature of action representation (from low-level motor commands to high-level task plans). This taxonomy provides a structured lens through which to analyze and compare diverse approaches, revealing underlying design principles and trade-offs. Second, we conduct a critical analysis of the state-of-the-art, examining how various VLA models [2, 3, 4, 5, 7, 8, 9, 10, 11] address (or fail to address) the core requirements of world modeling. We highlight key technical innovations, such as action expert distillation for efficient teaching [2510.09607v2], event-augmented architectures for robust perception [2604.04834v1], and adapter-based paradigms for resource-constrained settings [2509.09372v2], and assess their contributions to the WAM vision. Third, we identify and articulate the major open challenges and future research directions for realizing truly unified world action models. These include the need for scalable and generalizable world models, methods for handling counterfactual and out-of-distribution scenarios [2602.17659v1], techniques for real-time inference [2510.26742v1], and strategies for integrating diverse modalities and action spaces.

The significance of this survey lies in its timely synthesis of a rapidly evolving field. As embodied AI systems are increasingly deployed in unstructured, human-centric environments, the demand for agents that can both act and understand the consequences of their actions becomes paramount. By providing a clear conceptual framework and a comprehensive overview of the current landscape, this work aims to guide researchers and practitioners toward the next generation of intelligent, adaptable, and robust robotic systems. The motivation is clear: to move beyond reactive policies and toward a future where machines not only see and speak, but truly understand and anticipate the world they inhabit.

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The emergence of Vision-Language-Action (VLA) models represents a significant paradigm shift in embodied intelligence, aiming to unify visual perception, linguistic reasoning, and motor control within a single end-to-end framework. This section situates the proposed World Action Model (WAM) within the broader landscape of related work, organizing the discussion around three thematic axes: the architectural evolution and training strategies of VLA models, the integration of world modeling capabilities, and the critical challenges of robustness, efficiency, and generalization. By synthesizing findings across these areas, we identify a conspicuous gap

in the literature—the lack of a unified framework that seamlessly combines VLA with explicit world modeling—which the WAM directly addresses.

****Architectural Evolution and Training Strategies for VLA Models****

The foundational premise of VLA models is to leverage pre-trained vision-language models (VLMs) as a backbone for robotic control, thereby inheriting rich semantic understanding and commonsense reasoning. Early efforts in this direction demonstrated that fine-tuning VLMs on action prediction tasks could yield effective policies for manipulation and navigation. However, a major obstacle has been the computational and data cost associated with training large-scale VLAs from scratch. To mitigate this, several works have proposed efficient adaptation paradigms. For instance, VLA-Adapter introduces a lightweight adapter module that enables the transfer of a pre-trained VLM to robotic tasks with minimal parameter updates, making it particularly suitable for tiny-scale models and resource-constrained settings \cite{2509.09372v2}. Similarly, VITA-VLA employs an action expert distillation framework, wherein a pre-trained VLM is taught to act by distilling knowledge from a specialized action expert, thereby avoiding the need for large-scale robot interaction data \cite{2510.09607v2}. These approaches underscore a trend toward parameter-efficient fine-tuning, which preserves the general knowledge of the VLM while specializing it for embodied tasks.

A parallel line of research has explored the integration of motion priors and temporal dynamics into VLA architectures. Robotic VLA Benefits from Joint Learning with Motion Image Diffusion proposes to jointly train a VLA model with a motion image diffusion module, which generates future motion trajectories as visual representations \cite{2512.18007v1}. This joint learning paradigm is shown to improve both the quality of action predictions and the model's ability to reason about future states. This work highlights a crucial insight: purely static perception-action mappings are insufficient for complex, dynamic environments; incorporating predictive modeling of motion can enhance both performance and sample efficiency. The World Action Model builds on this insight by embedding a full world model that predicts future visual and state transitions, rather than merely motion images.

Another notable architectural innovation is the use of mixture-of-experts (MoE) to handle diverse manipulation skills. Towards Human-Like Manipulation through RL-Augmented Teleoperation and Mixture-of-Dexterous-Experts VLA introduces a framework where multiple specialized "dexterous experts" are trained via reinforcement learning and then combined through a gating mechanism within a VLA \cite{2603.08122v1}. This allows the model to exhibit human-like dexterity across a wide range of manipulation tasks. The MoE approach is particularly relevant to our work, as it provides a method for scaling VLA models to handle heterogeneous action spaces and task requirements. However, these expert modules are

typically static and do not incorporate a dynamic world model for online planning.

****Integration of World Modeling and Predictive Capabilities****

World modeling—the ability to simulate the consequences of actions and reason about future states—has long been recognized as a cornerstone of intelligent behavior in robotics. While traditional model-based reinforcement learning has extensively explored this area, its integration with VLA models remains nascent. The LongNav-R1 system addresses long-horizon navigation by employing horizon-adaptive multi-turn reinforcement learning, which allows the VLA to adjust its planning horizon based on the task context \cite{2602.12351v1}. Although not a full world model, this adaptive horizon mechanism implicitly requires the agent to predict future states over varying time scales, thus bridging the gap between reactive control and deliberative planning.

More explicitly, Impromptu VLA provides an open-weight and open-data framework for driving VLAs, emphasizing the importance of transparency and reproducibility in this emerging field \cite{2505.23757v1}. While primarily focused on the driving domain, the work underscores the need for models that can predict the behavior of other traffic participants and the evolution of the scene—a form of world modeling. The authors demonstrate that training on large-scale, diverse driving data enables the VLA to generalize to novel scenarios, but they do not incorporate an explicit, structured world model.

The Event-Augmented VLA (E-VLA) model takes a different approach to robustness by integrating event-based cameras to handle dark and blurred scenes \cite{2604.04834v1}. By augmenting the visual input with high-temporal-resolution event data, the model becomes more resilient to perceptual failures that plague conventional RGB-based VLAs. This work is significant because it addresses a key limitation of world models: their reliance on high-quality visual observations. However, E-VLA focuses on input modality fusion rather than on learning a predictive model of the world.

A pragmatic view of VLA foundation models is presented in A Pragmatic VLA Foundation Model, which argues for a task-agnostic, large-scale pre-training approach that yields a general-purpose action foundation model \cite{2601.18692v2}. This model is trained on a diverse corpus of robotic demonstrations and is shown to generalize to unseen tasks and embodiments. While impressive in scale, this work does not explicitly model the dynamics of the environment; instead, it relies on the implicit statistical regularities in the training data to capture action consequences. The World Action Model differs fundamentally by incorporating an explicit, learnable world model that predicts future observations and rewards, enabling both planning and counterfactual reasoning.

****Challenges of Robustness, Efficiency, and Generalization****

Despite rapid progress, VLA models face several critical challenges that limit their real-world deployment. One prominent issue is robustness to distribution shift and counterfactual scenarios. When Vision Overrides Language systematically evaluates VLA failures in situations where visual information contradicts linguistic instructions \cite{2602.17659v1}. The authors find that VLAs often default to visual cues, ignoring the language command, leading to catastrophic failures. They propose a mitigation strategy based on adversarial training and contrastive learning. This work highlights a fundamental limitation of current VLAs: they lack a world model that can simulate counterfactual outcomes and thus fail to resolve conflicts between modalities. The World Action Model, by maintaining an internal model of state transitions, can evaluate the consequences of following either the visual or linguistic modality, thereby enabling more robust decision-making.

Efficiency is another major concern, particularly for real-time deployment on resource-constrained robots. Running VLAs at Real-time Speed addresses this by proposing a series of optimizations, including model quantization, architectural pruning, and efficient attention mechanisms \cite{2510.26742v1}. While these techniques are essential for practical deployment, they often come at the cost of model expressiveness and predictive accuracy. Our proposed framework complements these efforts by demonstrating that a world model can be learned and queried efficiently, reducing the need for large, reactive neural networks.

Finally, the challenge of data scarcity and task diversity remains acute. Most VLA models are trained on a limited set of tasks and environments, hindering generalization. The Robotic VLA Benefits from Joint Learning with Motion Image Diffusion study shows that joint training with generative models can alleviate data requirements by providing synthetic training signals \cite{2512.18007v1}. Similarly, VLA-Adapter \cite{2509.09372v2} and VITA-VLA \cite{2510.09607v2} demonstrate that distillation and fine-tuning can reduce the need for large-scale robot data. However, these approaches still require task-specific demonstrations. The World Action Model, by learning a general world model from diverse interaction data, can in principle be applied to novel tasks through planning within the learned model, without requiring task-specific demonstrations.

****Summary and Identified Gap****

In summary, the literature on VLA models has advanced along three main fronts: architectural innovation for efficient adaptation, integration of predictive and world modeling elements, and addressing robustness and efficiency challenges. However, existing approaches treat world

modeling as either an implicit byproduct of large-scale training \cite{2601.18692v2}, a specialized module for motion prediction \cite{2512.18007v1}, or an adaptive planning horizon \cite{2602.12351v1}. No prior work has proposed a unified framework that explicitly integrates a full, differentiable world model with a VLA architecture in a manner that enables online planning, counterfactual reasoning, and robust generalization across tasks and embodiments. The World Action Model fills this gap by providing a cohesive paradigm where perception, language understanding, action generation, and world prediction are jointly learned and synergistically combined. This unification not only enhances performance on standard benchmarks but also opens new avenues for interpretability, safety, and sample efficiency in embodied intelligence.

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This review has systematically examined the emerging paradigm of World Action Models (WAMs), a novel framework that seeks to unify Vision-Language-Action (VLA) models with world modeling for embodied intelligence. The central thesis advanced is that while VLA models have demonstrated remarkable progress in grounding language instructions into physical actions, their operational scope remains fundamentally limited by a lack of intrinsic world understanding—an understanding of causality, physics, and long-term dynamics that is essential for robust, generalizable, and safe deployment in unstructured environments.

The key findings of this survey are threefold. First, we have established that current VLA architectures, despite their successes in tasks such as robotic manipulation \cite{2603.08122v1} and navigation \cite{2602.12351v1}, predominantly function as reactive policies. They map perceptual inputs and linguistic commands directly to motor outputs without an internal model of how the world evolves. This limitation manifests in critical failure modes, including sensitivity to counterfactual scenarios where visual input conflicts with linguistic priors \cite{2602.17659v1}, and degraded performance under non-ideal conditions such as low lighting or motion blur \cite{2604.04834v1}. Second, we have identified the core components of a unified WAM architecture: a world model that predicts future states, a policy that acts within that predicted world, and a perceptual encoder that bridges raw sensory data with abstract representations. The reviewed literature suggests that joint training of these components—for instance, by learning action policies alongside motion image diffusion \cite{2512.18007v1} or

through expert distillation that teaches models to anticipate consequences \cite{2510.09607v2}—yields synergistic benefits that neither paradigm achieves in isolation. Third, we have documented the emergence of pragmatic engineering solutions that facilitate real-world deployment, including real-time inference frameworks \cite{2510.26742v1}, lightweight adapter modules for tiny-scale models \cite{2509.09372v2}, and open-weight driving-specific implementations \cite{2505.23757v1}.

The principal contributions of this work are threefold. First, it provides the first comprehensive taxonomy and critical analysis of the WAM paradigm, situating it within the broader trajectory of embodied AI research. Second, it synthesizes disparate lines of inquiry—ranging from foundational VLA architectures \cite{2601.18692v2} to domain-specific adaptations for autonomous driving \cite{2505.23757v1} and dexterous manipulation \cite{2603.08122v1}—into a coherent theoretical framework. Third, it identifies the critical bottlenecks that must be addressed for WAMs to achieve their full potential, including the need for scalable world model training, robust evaluation protocols that test causal reasoning, and efficient architectures that maintain predictive fidelity under real-time constraints.

Several promising directions for future research emerge from this synthesis. A primary avenue is the development of hybrid architectures that combine the reactive efficiency of VLA policies with the predictive foresight of learned world models, perhaps through differentiable planning loops or latent imagination mechanisms. Another critical direction is the creation of large-scale, diverse benchmarks that explicitly test a model's ability to reason about physical dynamics and long-horizon consequences, moving beyond the current focus on short-term task completion. Furthermore, the integration of event-based sensing \cite{2604.04834v1} and multi-modal fusion techniques could enhance robustness in challenging perceptual conditions. Finally, the ethical and safety implications of deploying WAMs in human environments warrant rigorous investigation, particularly concerning the predictability and interpretability of actions generated by models that internally simulate future states.

In conclusion, the unification of vision-language-action models with world modeling represents a foundational shift toward genuinely intelligent embodied systems. By moving beyond reactive policies to agents that can reason, plan, and anticipate, the WAM paradigm holds the potential to unlock a new generation of robots that are not merely tools but collaborative partners capable of adaptive, safe, and general-purpose interaction with the physical world. The path forward demands sustained interdisciplinary effort, but the promise is nothing less than a machine that truly understands the world it inhabits.

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